

Reconstructing frustrated signs from amplitudes with fixed-node and Krylov updates

A. Otaifi¹

¹*Arnold Sommerfeld Center for Theoretical Physics,
Ludwig-Maximilians-Universität München, 80333 München, Germany*

(Dated: October 1, 2026)

We separate the sign and amplitude problems of a frustrated wave function, $\psi(x) = s(x)a(x)$, in the square-lattice spin-1/2 J_1 - J_2 Heisenberg model. For fixed amplitude, one shifted Krylov step gives $s'(x) = s(x) \text{sgn}[T - r_s(x)]$, where $r_s = (H\psi)/\psi$. At $J_2/J_1 = 0.5$ this reduces the exact 4×4 wrong-sign weight from 1.27×10^{-2} to 3.41×10^{-4} . On 6×6 , with the same neural-network amplitude, the reconstructed signs are energetically indistinguishable from the original neural signs. Conversely, fixed-node projection supplies a sign-free amplitude update; an exact 4×4 fixed-node/Krylov loop reaches the exact ground-state signs. On 8×8 , the reconstructed signs retain a competitive fixed-node energy. The remaining scaling problem is learning a callable amplitude from fixed-node walkers. Maximum likelihood targets $a_{\text{new}} \propto \sqrt{a\phi_{\text{FN}}}$, and stochastic reconfiguration yields an update direction that transfers between independent walker populations.

I. INTRODUCTION

Frustration produces sign-changing path weights and the Monte Carlo sign problem [1, 2]. The square-lattice J_1 - J_2 Heisenberg model is a standard case: the Marshall sign rule is exact at $J_2 = 0$ [3], but fails near $J_2/J_1 = 1/2$, where accurate variational states are already nontrivial [4, 5].

We ask whether signs and amplitudes can be improved separately. Write a real wave function as

$$\psi(x) = s(x)a(x), \quad s(x) = \pm 1, \text{quad}a(x) > 0. \quad (1)$$

Given a , a single Krylov step changes only s . Given s , lattice fixed node gives a sign-free positive-amplitude problem [6]. One iteration is therefore

We test these two operations separately and then together. Exact diagonalization establishes the sign update, 6×6 and 8×8 calculations test its physical effect, and an exact 4×4 loop tests closure. The large-system bottleneck is then isolated to the amplitude handoff from fixed-node walkers.

II. MODEL AND METHOD

We study

$$H = J_1 \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + J_2 \sum_{\langle\langle ij \rangle\rangle} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (2)$$

on periodic $L \times L$ lattices with $J_1 = 1$ and $S^z = 0$.

A. Krylov sign update

For $\psi = sa$, define the local energy

$$r_s(x) = \frac{(H\psi)(x)}{\psi(x)}. \quad (3)$$

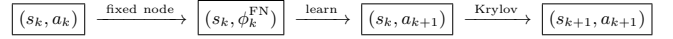


FIG. 1. One feedback iteration. Fixed node updates the amplitude at fixed signs; the Krylov step then updates the signs.

Since

$$[(T - H)\psi](x) = [T - r_s(x)]\psi(x), \quad (4)$$

the sign of the shifted Krylov vector is

$$s_K(x) = s(x) \text{sgn}[T - r_s(x)]. \quad (5)$$

We keep this sign and discard the Krylov amplitude factor $|T - r_s|$. Thus the method is not repeated Lanczos multiplication. For exact systems, T minimizes the fixed-amplitude energy; on sampled systems it is obtained from a two-cluster split of the r_s distribution.

When the exact ground state $\psi_0 = s_0 a_0$ is available, we report

$$O_s = \left| \sum_x a_0(x)^2 s(x) s_0(x) \right|, \text{quad}w_s = \frac{1 - O_s}{2}, \quad (6)$$

where w_s is the exact probability weight carrying the wrong sign.

B. Fixed-node amplitude update

For guide ψ_G , a bond $x \leftrightarrow y$ is sign violating when $H_{xy}\psi_G(y)/\psi_G(x) > 0$. Lattice fixed node removes these off-diagonal terms and adds their contribution to the diagonal [6]. The resulting Hamiltonian is sign free after importance sampling and has the imposed guide signs.

If its positive ground-state modulus is ϕ_{FN} , an exact small-system loop can feed ϕ_{FN} directly into Eq. (5). In a walker calculation the stationary mixed distribution is instead

$$f(x) \propto a(x)\phi_{\text{FN}}(x). \quad (7)$$

For a Born model $p_\theta(x) \propto a_\theta(x)^2$, exact maximum likelihood therefore gives

$$a_{\text{new}}(x) \propto \sqrt{a(x)\phi_{\text{FN}}(x)}. \quad (8)$$

The 4×4 exact calculations use the full $S^z = 0$ space (12 870 states). For 6×6 and 8×8 we use a previously optimized translationally invariant vision-transformer neural quantum state (ViT) at $J_2/J_1 = 0.5$. The controlled sign tests hold its modulus fixed. The 8×8 fixed-node runs use population $M = 128$ and projection length $\beta = 1.2$.

III. ONE-STEP SIGN RECONSTRUCTION

Figure 2(a,b) uses the exact 4×4 ground-state modulus. At $J_2/J_1 = 0.5$, Marshall signs have $O_s = 0.974538$ ($w_s = 1.27 \times 10^{-2}$). One Krylov update gives $O_s = 0.999319$ ($w_s = 3.41 \times 10^{-4}$) and reduces the fixed-amplitude energy error per site from 1.16×10^{-2} to 5.51×10^{-4} . The same construction remains effective across the sweep when the parent sign pattern is chosen in the appropriate Marshall or stripe basin.

The 6×6 test removes the exact-sign oracle. Using the same ViT modulus for all states,

$$\begin{aligned} e_{\text{Marshall}} &= -0.4981534, \\ e_{\text{Krylov}} &= -0.5033337, \\ e_{\text{ViT}} &= -0.5034504. \end{aligned} \quad (9)$$

The paired difference $e_{\text{Krylov}} - e_{\text{ViT}} = 1.17(287) \times 10^{-4}$ is unresolved, while Krylov improves over Marshall by 4.79×10^{-3} per site. Thus the energy gain comes from the signs, not from changing the amplitude.

The mechanism is visible in the Marshall local field,

$$r_M = D + F_{J_1} + F_{J_2}. \quad (10)$$

The useful signal is mainly the competition between the two off-diagonal terms. On independent 6×6 samples, the physically weighted wrong-sign mass falls from 0.01635 to 8.08×10^{-4} (train) and from 0.01811 to 8.02×10^{-4} (validation), i.e. about a twenty-fold reduction.

IV. CLOSING THE LOOP EXACTLY

We now target $J_2/J_1 = 0.5$ but initialize with the exact $J_2 = 0$ modulus and Marshall signs. At each step we solve the current fixed-node problem exactly, replace the amplitude by its positive modulus, and recompute Eq. (5). Target-state information is used only for diagnostics.

Figure 3 shows the result. The initial amplitude fidelity is 0.7132 and the sign overlap is 0.974538. After two iterations the sign overlap is 0.995945, after twenty it

is 0.9999982, and at iteration 100 the exact sign pattern is reached. The final amplitude fidelity is 0.9999946 and the guide energy is 2.60×10^{-5} above the exact ground state.

The plateaus are physical: fixed node can improve the amplitude for many steps without changing a sign. A sign update occurs only when the improved amplitude shifts the local-energy threshold across a new set of configurations.

V. SCALING TO 8×8

At $M = 128$, the two-population means are

$$E_{\text{K-FN}} = -31.88542, \quad E_{\text{M-FN}} = -31.67748. \quad (11)$$

Their difference is -0.20795 . Two-population spreads are used only as reproducibility scales.

The Krylov-sign mean is $E/N = -0.498210$, compared with -0.494961 for Marshall signs. Figure 4 shows that it lies in the same energy range as high-accuracy DMRG/FAMPS and neural benchmarks on the same geometry. We use this only as a scale comparison, not as a ranking between methods.

The sign update therefore survives to 8×8 . The bottleneck is now different: fixed-node walkers must be converted into an amplitude that can be evaluated on arbitrary configurations.

VI. LEARNING THE AMPLITUDE FROM WALKERS

Equation (8) can be tested exactly on 4×4 . Starting from the exact target modulus with Marshall signs, one ideal half-step plus the Krylov sign update changes the energy from -8.27291 to -8.44579 , close to $E_0 = -8.45792$. A uniform-amplitude start is initially non-monotone but subsequently recovers [Fig. 5(a)].

At 8×8 , the issue is sampling noise. Let g_1 and g_2 be maximum-likelihood gradients from two independent fixed-node populations. Their Euclidean cosine is

$$\frac{g_1 \cdot g_2}{\|g_1\| \|g_2\|} = -0.853, \quad (12)$$

so an ordinary gradient step does not transfer between populations.

We therefore use stochastic reconfiguration (SR) [8],

$$\delta\theta = (S + \lambda I)^{-1} g_1. \quad (13)$$

For $\lambda = 1$, the held-out directional derivative becomes $g_2^\top \delta\theta = +3.85$. The finite line search in Fig. 5(b) remains positive and peaks near $\eta = 0.01$, where the held-out log-likelihood gain is 0.0156 and the reweighting effective sample size is 3236/4096.

This establishes a transferable amplitude-learning direction. It does not yet establish convergence of a repeatedly learned 8×8 loop; fixed-node energy remains the physical acceptance test.

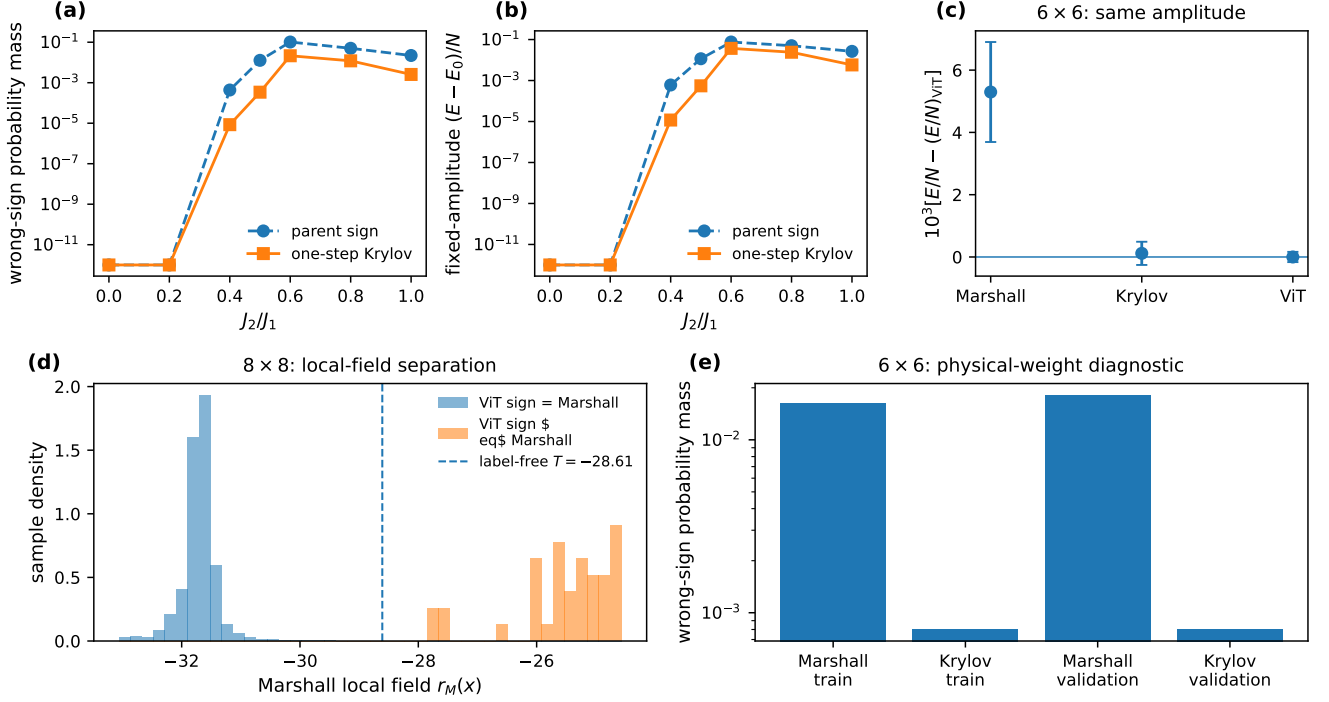


FIG. 2. Sign reconstruction. (a,b) Exact 4×4 sign weight and fixed-amplitude energy error. (c) 6×6 energies with identical ViT modulus and different signs. (d) 8×8 Marshall local-field distributions, colored afterward by the ViT sign. (e) Independent 6×6 train/validation wrong-sign weights.

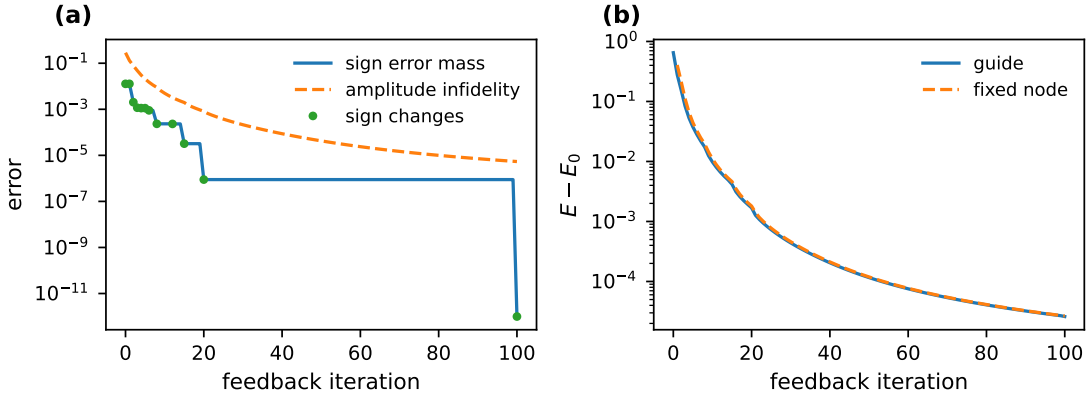


FIG. 3. Exact 4×4 fixed-node/Krylov loop at $J_2/J_1 = 0.5$. (a) Sign error and amplitude infidelity. Markers denote sign changes. (b) Guide and fixed-node energy errors.

VII. DISCUSSION AND OUTLOOK

The data point to a simple division of labor. A good amplitude makes the dominant frustrated sign correction almost one-dimensional: a threshold in the local Krylov field. Fixed node solves the complementary positive-amplitude problem at fixed signs, and the exact 4×4 loop shows that the two updates can close.

The unresolved large-system problem is the amplitude handoff from projector walkers. SR makes one 8×8 update transferable across independent populations, but re-

peated updates still need fixed-node energy validation. The method also requires a useful initial amplitude and should not be confused with repeated bare Krylov iteration; the direct second step is worse on 6×6 .

Appendix A: Direct second Krylov step

On 6×6 , retaining the full first Krylov amplitude gives $a_1(x) = |T_1 - r_0(x)|a_0(x)$ with coefficient of variation about 0.18. A second unsupervised threshold changes

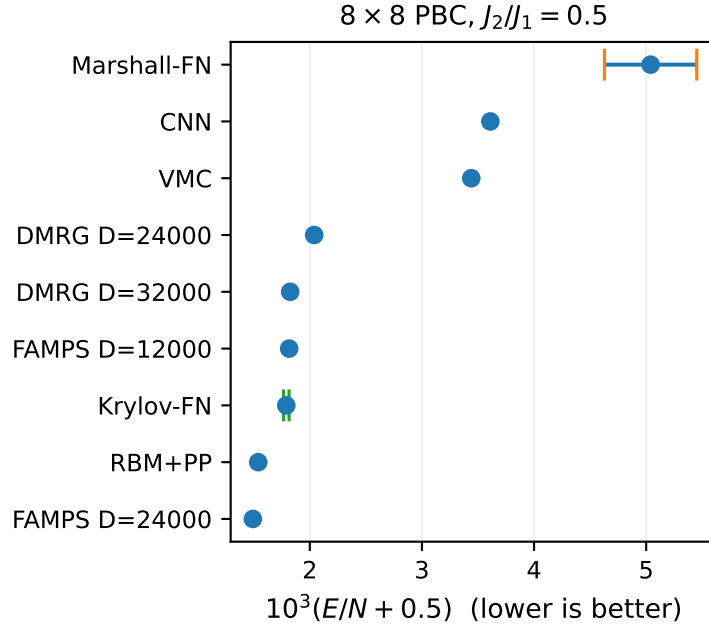


FIG. 4. 8×8 periodic-boundary energy benchmark at $J_2/J_1 = 0.5$. Literature values are from Ref. [7]; ticks mark the two fixed-node populations.

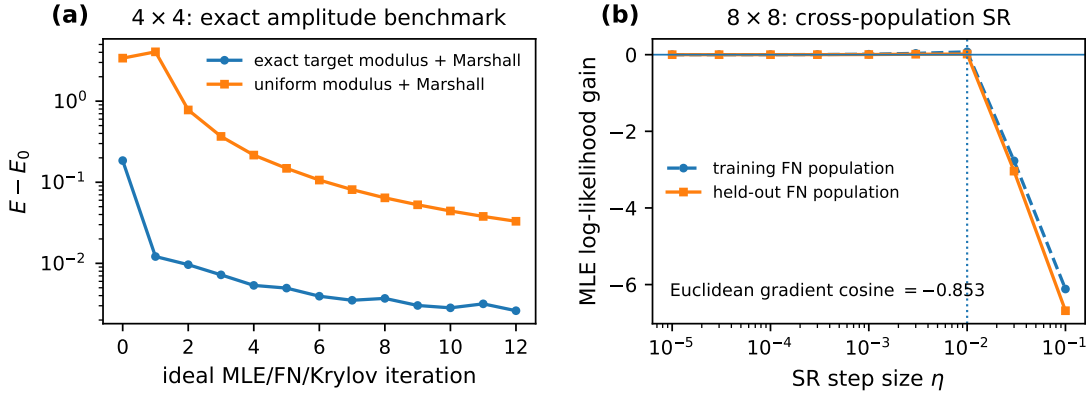


FIG. 5. Amplitude handoff. (a) Exact 4×4 iterations of the ideal MLE half step, Eq. (8). (b) 8×8 SR line search: the update is fitted on one fixed-node population and tested on another.

only 0.8–2.0% of physical weight but raises the fixed-amplitude energy by 0.65–0.96 in total energy. The feedback construction therefore restores or re-solves the amplitude before another sign update.

Appendix B: Statistical convention

Exact-diagonalization data are deterministic up to eigensolver precision. For the 6×6 fixed-amplitude comparison we use correlation-robust between-chain standard errors. For 8×8 fixed-node data with two independent populations, error bars denote half the population difference and are treated only as reproducibility scales.

[1] P. Henelius and A. W. Sandvik, Sign problem in monte carlo simulations of frustrated quantum spin systems,

Phys. Rev. B **62**, 1102 (2000).

- [2] M. Troyer and U.-J. Wiese, Computational complexity and fundamental limitations to fermionic quantum monte carlo simulations, *Phys. Rev. Lett.* **94**, 170201 (2005).
- [3] W. Marshall, Antiferromagnetism, *Proc. R. Soc. Lond. A* **232**, 48 (1955).
- [4] K. Choo, T. Neupert, and G. Carleo, Two-dimensional frustrated j_1 - j_2 model studied with neural network quantum states, *Phys. Rev. B* **100**, 125124 (2019).
- [5] A. Chen and M. Heyl, Empowering deep neural quantum states through efficient optimization, *Nat. Phys.* **20**, 1476 (2024).
- [6] H. J. M. van Bommel, D. F. B. ten Haaf, W. van Saarloos, J. M. J. van Leeuwen, and G. An, Fixed-node quantum monte carlo method for lattice fermions, *Phys. Rev. Lett.* **72**, 2442 (1994).
- [7] X. Qian and M. Qin, Augmenting density matrix renormalization group with disentanglers, *Chin. Phys. Lett.* **40**, 057102 (2023).
- [8] S. Sorella, Generalized lanczos algorithm for variational quantum monte carlo, *Phys. Rev. B* **64**, 024512 (2001).